

Corporate Announcements, Price, and Volume: An Event Study of Five NSE Stocks

Quantitative Data Analyst Intern Assessment - Final Report

Executive Summary

This project measures how corporate announcements relate to subsequent stock price and trading-volume behavior for RELIANCE, HDFCBANK, NYKAA, HAL, and RVNL. The supplied pack contains 2530 announcement rows and one-minute OHLCV bars. 2528 announcements align to a first complete one-minute bar at or after dissemination. 2 announcements are excluded because no later bar exists. Stage 2 groups the notices into nine subject categories and 2078 event clusters. Stage 3 computes close-to-close returns, a prior-session volume ratio, and a high-low range. Stage 4 runs price-conditioned PEAD on 75 independent financial-result events. This is not an earnings-surprise study.

The strongest descriptive patterns are concentrated activity on financial-result days (median 5-minute volume ratio 6.32, $n=74$), a slightly negative same-session financial-result return (mean -0.0050 , $n=75$), and later-session means that keep the sign of the event-day price reaction. None of these results is presented as a trading rule, a causal claim, or a statistically tested anomaly.

1. Objective

The assessment asks a practical market-microstructure question: after a company publishes an announcement, how do price and volume behave over short intraday windows and over the next several trading sessions? The required work includes timestamp alignment, event independence, a transparent subject taxonomy, event-level impact, subject-level comparison, and price-conditioned PEAD on financial-result events. The implementation stays close to the supplied files. No external prices, analyst estimates, or earnings-surprise databases are used.

2. Data

The assessment folder supplies corporate_announcements.csv (and a jsonl copy), metadata.json, and five one-minute market files: RELIANCE, HDFCBANK, NYKAA, HAL, and RVNL. DATA_FORMAT.md is named in the assignment and is not present in the pack. Raw files are not stored in the public project. A reviewer points --data-dir at the supplied folder or sets QUANT_DATA_DIR.

Item	Value
Announcement rows	2530
Aligned announcements	2528
Excluded announcements	2
Event clusters	2078
Stocks	RELIANCE, HDFCBANK, NYKAA, HAL, RVNL
Event trading-date span	2023-08-22 to 2026-08-19
Market sessions per stock (Stage 1)	744

Each market file has about 277,500 one-minute bars. Two invalid OHLC bars exist (HDFCBANK and HAL, both 2024-06-25 09:15). Zero-volume bars appear in every stock. Sessions are inferred from actual timestamps and include weekend and non-standard clocks. The study does not assume a Monday-Friday 09:15-15:30 calendar.

3. Data Quality and Timestamp Alignment

Announcement time prefers DisseminationDT when it parses. DT_TM is the fallback. Mixed ISO precision is accepted with format=ISO8601. Naive pd.to_datetime rejected 11 valid timestamps in early testing, so the ISO parser is required. Timestamps stay naive as supplied because DATA_FORMAT.md is missing and no timezone is invented.

The event bar is the first one-minute bar whose start is greater than or equal to the chosen announcement timestamp. A bar that began before dissemination is never used. After-close announcements map to the next available session. The two exclusions are HDFCBANK notices after the last supplied bar on 2026-08-19. Invalid OHLC rows are flagged and not repaired. Metrics that need those prices are marked unavailable.

4. Event Clustering and Subject Taxonomy

Related notices are clustered when they share a symbol and the same normalized NEWSSUB and are at most 48 hours apart. A financial-result notice may also merge with a nearby investor presentation or earnings transcript for the same symbol. Different subjects on the same day are not merged just because they share a date. `is_financial_result` is conservative: CATEGORYNAME Result or tight result phrases on a small set of categories. Analyst meets, transcripts, presentations, and board-meeting intimations are not treated as results. The nine subject groups come from documented if/then rules. A group is inferential when it has at least 10 independent events (unique `cluster_id` values). All nine groups meet that cutoff.

Subject group	Announcements	Clusters	Status
board_governance	367	352	inferential
business_updates	723	688	inferential
capital_structure	577	300	inferential
corporate_actions	32	32	inferential
credit_ratings	34	32	inferential
financial_results	93	75	inferential
investor_communication	421	370	inferential
regulatory_compliance	129	128	inferential
strategic_transactions	154	129	inferential

Independent-event counts can exceed unique `cluster_id` values because a cluster may contain more than one subject after the financial-result / presentation merge. Subject-level analysis therefore uses one observation per (`subject_group`, `cluster_id`).

5. Event-Level Price, Volume, and Range Analysis

The impact anchor is `effective_bar_timestamp`. The return is `future_close / anchor_close - 1`. Intraday 5m, 30m, and 60m horizons use the first same-session bar at or after anchor plus N minutes. If the session ends first, the metric is missing. Session close uses the last actual bar of that session, not an assumed 15:30 clock. D+1, D+5, D+10, and D+20 are inferred trading sessions, not calendar days.

Volume ratio equals event-window volume divided by the median of the same clock-time window in prior sessions. The lookback is at most 20 prior sessions and needs at least 5 usable windows. The event session and later sessions are excluded. Range is `window_high / window_low - 1`. Zero-volume bars stay in the file and can still contribute valid prices.

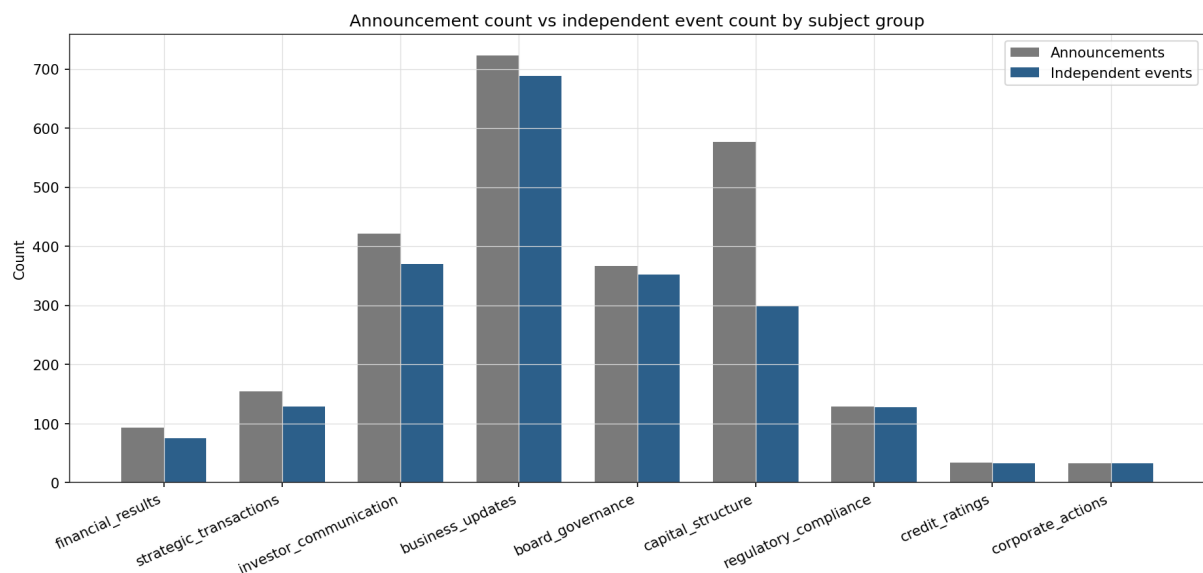


Figure 1. Announcement count is not the independent sample. capital_structure is the clearest gap.

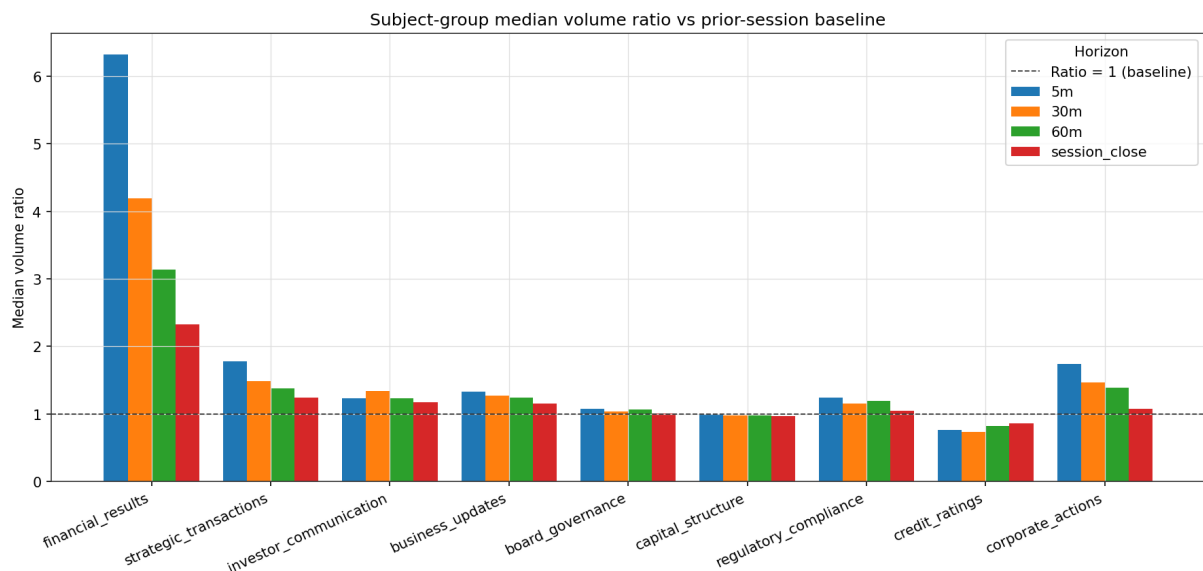


Figure 2. Median volume ratio by subject. A value of 1 matches the prior-session clock-time baseline.

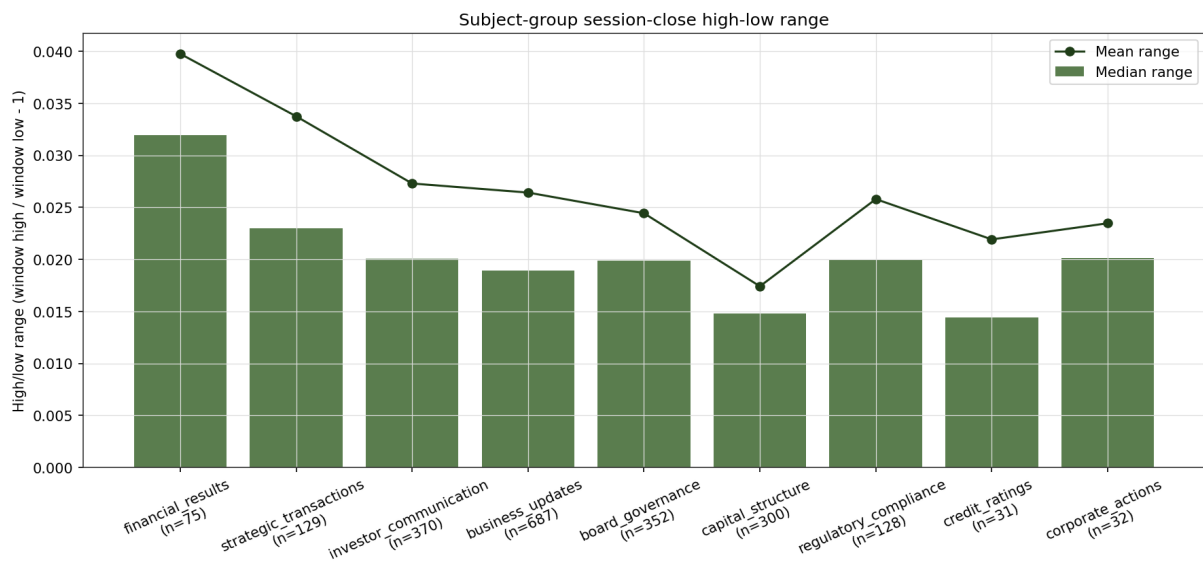


Figure 3. Session-close high-low range by subject group.

6. Subject-Level Results

Subject summaries use independent events. Uncertainty is a normal-approximation 95% interval: mean $\pm 1.96 \times \text{std} / \sqrt{n}$. Intervals are unavailable when $n < 2$. Groups with fewer than 10 independent events would be marked descriptive_only. At the pooled subject level all nine groups are inferential. Stock-subject cells are often descriptive_only and are retained, not deleted.

Subject	n	Mean SC	Median SC	Mean D+5	Median D+5
board_governance	352	0.0002	-0.0007	0.0010	0.0003
business_updates	687	-0.0018	-0.0013	0.0033	0.0012
capital_structure	300	-0.0010	-0.0009	0.0006	0.0014
corporate_actions	32	-0.0029	-0.0025	0.0176	0.0002
credit_ratings	31	0.0039	0.0024	0.0034	0.0002
financial_results	75	-0.0050	-0.0063	0.0047	0.0014
investor_communication	370	-0.0021	-0.0020	0.0066	0.0004
regulatory_compliance	128	0.0003	-0.0006	0.0000	-0.0062
strategic_transactions	129	0.0011	-0.0017	0.0097	0.0052

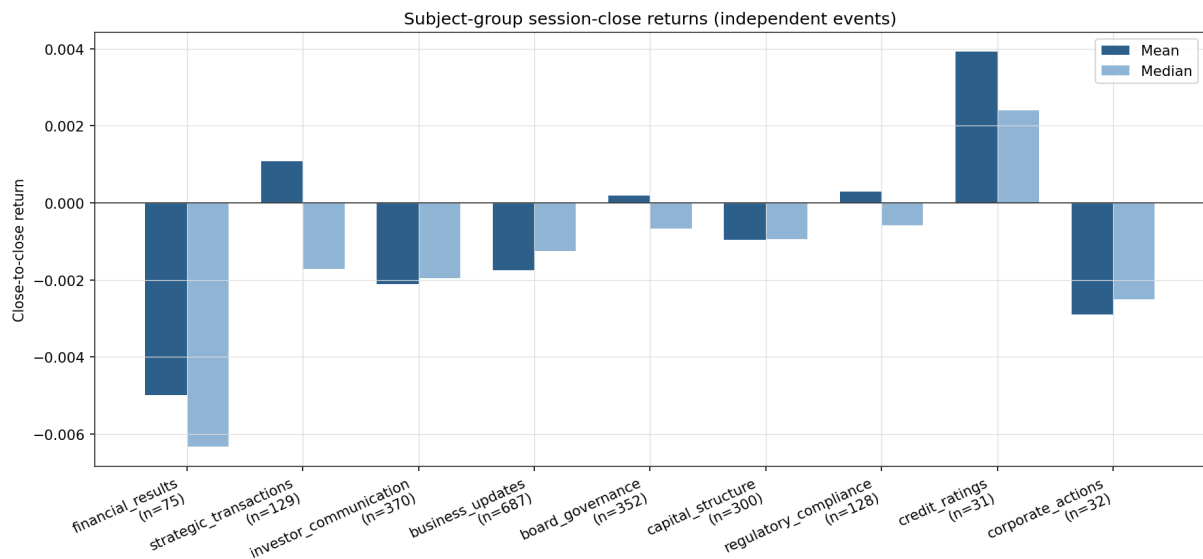


Figure 4. Session-close mean and median returns by subject. Sample sizes are on the axis labels.

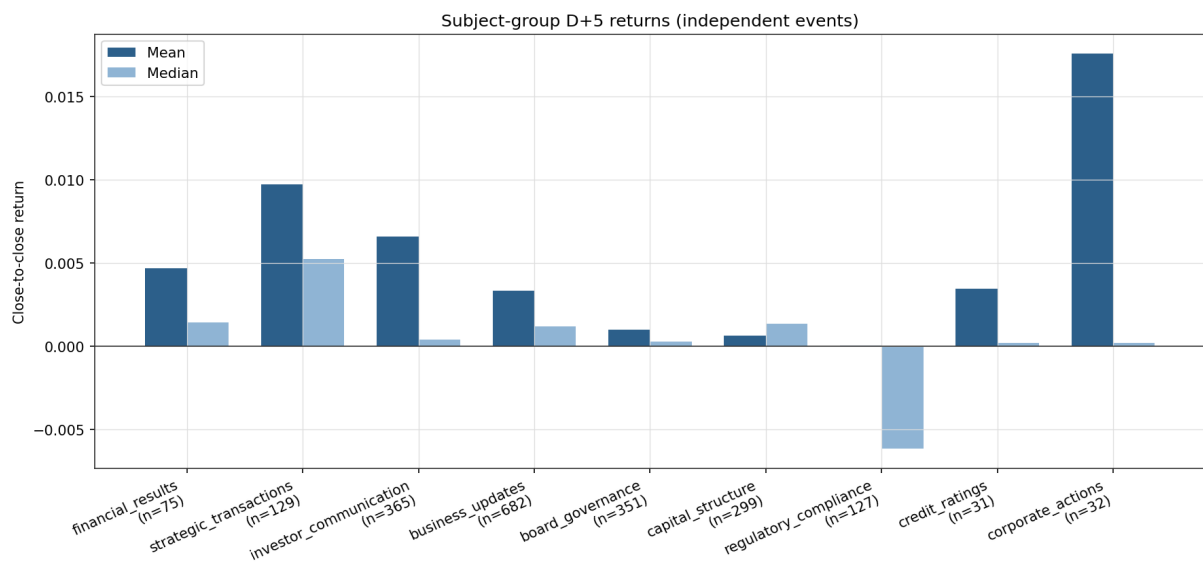


Figure 5. D+5 mean and median returns. corporate_actions has a large mean-median gap and is not treated as a stable pattern.

Financial-result days stand out on volume and range more than on pooled same-session return. business_updates, the largest residual group, has a small negative session-close mean and a tight interval. That group mixes many notice types, so the mean is a description of a mixed bag, not one mechanism. 60-minute missingness is highest for late-session events and reaches about 10% for regulatory_compliance.

7. Price-Conditioned PEAD

This analysis is price-conditioned and does not use earnings-surprise data. The sample is one aligned financial-result event per cluster (earliest effective bar). Initial reaction is the Stage 3 session-close return: positive if greater than 0, neutral if exactly 0, negative if less than 0. The sample has 26 positive, 0 neutral, and 49 negative events. Horizons are D+1, D+3, D+5, D+10, and D+20 inferred sessions.

Reaction	Horizon	n	Mean	Median	CI low	CI high
all	d1	75	-0.0094	-0.0046	-0.0190	0.0002
all	d3	75	0.0023	-0.0052	-0.0122	0.0167
all	d5	75	0.0047	0.0014	-0.0104	0.0197
all	d10	72	0.0005	-0.0110	-0.0186	0.0197
all	d20	71	0.0128	0.0040	-0.0099	0.0355
positive	d1	26	0.0139	0.0070	0.0048	0.0230
positive	d3	26	0.0381	0.0225	0.0198	0.0563
positive	d5	26	0.0357	0.0243	0.0191	0.0523
positive	d10	24	0.0172	-0.0003	-0.0055	0.0399
positive	d20	24	0.0281	0.0334	0.0013	0.0549
negative	d1	49	-0.0217	-0.0219	-0.0344	-0.0091
negative	d3	49	-0.0167	-0.0175	-0.0345	0.0011
negative	d5	49	-0.0118	-0.0185	-0.0317	0.0081
negative	d10	48	-0.0078	-0.0260	-0.0340	0.0184
negative	d20	47	0.0051	-0.0169	-0.0263	0.0365

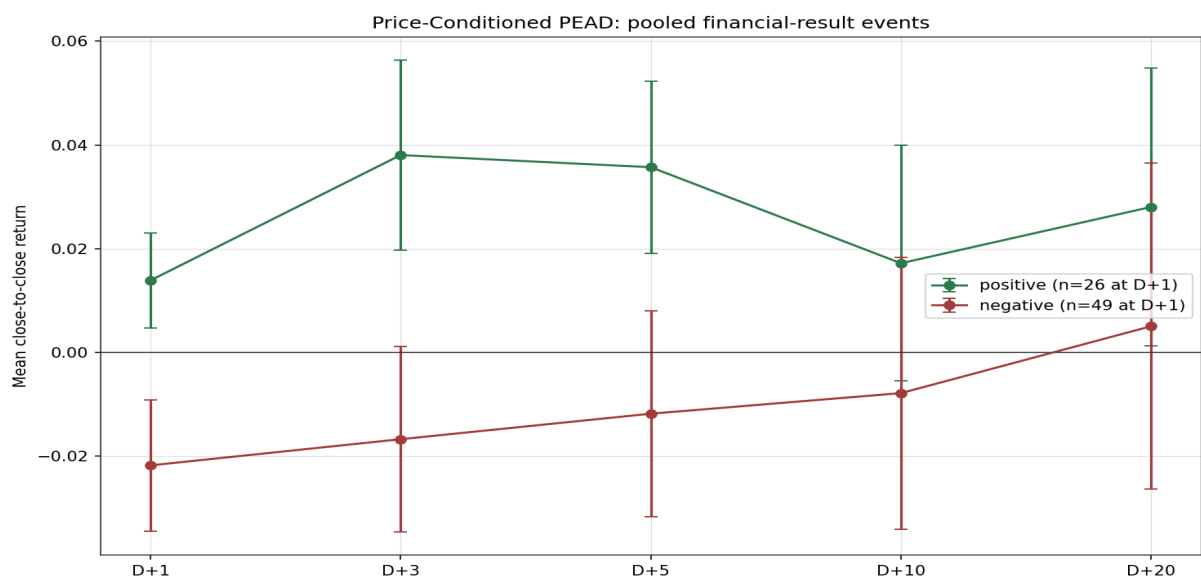


Figure 6. Price-Conditioned PEAD. Neutral initial-reaction events had zero observations.

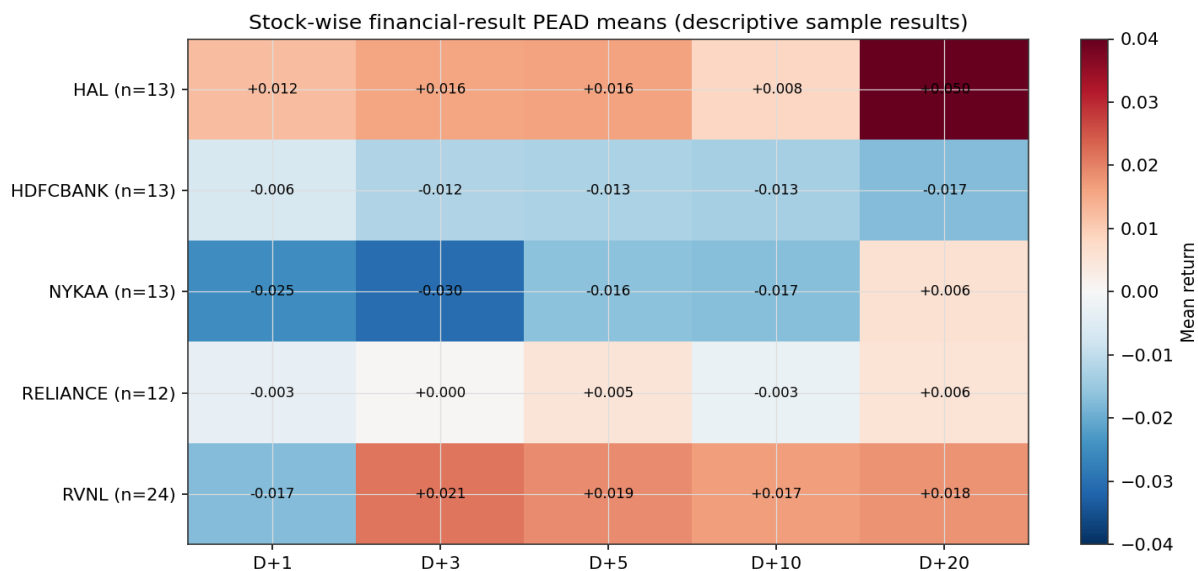


Figure 7. Stock-wise financial-result means. These are descriptive sample results, not stock rankings.

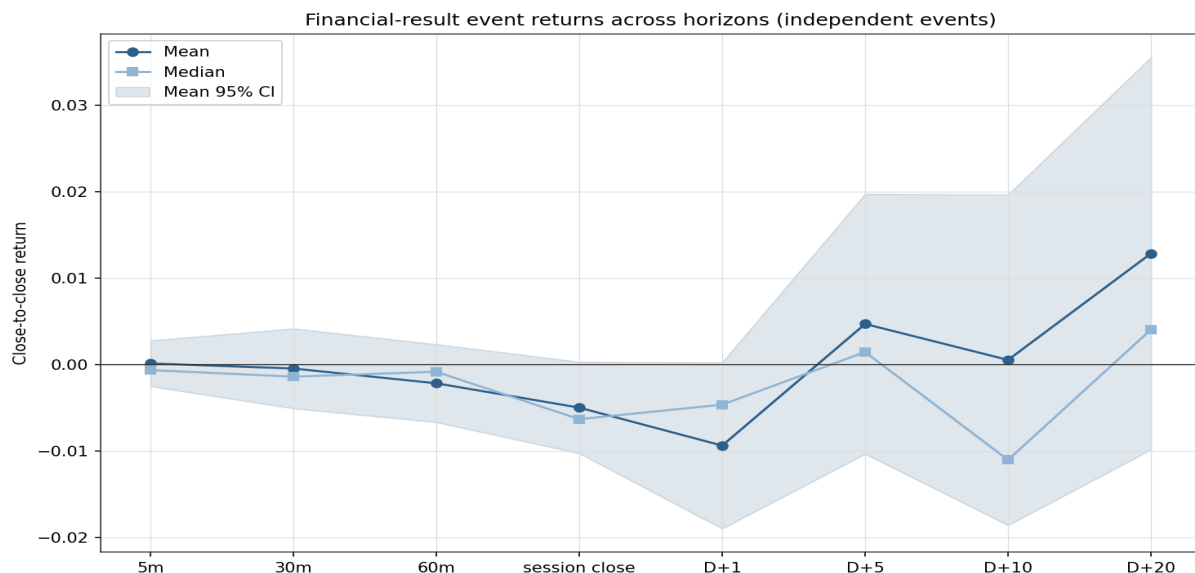


Figure 8. Independent financial-result returns from 5 minutes through D+20.

The unconditioned D+1 mean is -0.0094 (n=75). After splitting on the event-day sign, later-session means keep that sign in the pooled sample. Stock-wise sign cells are mostly smaller than 10 observations and are marked descriptive. No market-model residual is removed.

8. Key Insights

Each insight states an observation, a magnitude, a sample size, a cautious interpretation, and a limitation. Rejected candidates included the corporate_actions D+5 mean, which is not supported by the median, and any wording that would treat PEAD as an earnings-surprise anomaly.

11. financial results volume

Observation. Among 75 independent financial_results events, the median 5-minute volume ratio was 6.32 (mean 15.57, n=74), compared with 0.99 for capital_structure.

Why it matters. Financial-result windows show more trading activity than a ratio of 1, which is the prior-session clock-matched baseline.

Limitation. The mean is larger than the median, so a few high-volume events pull the average. This is a volume comparison, not a return forecast.

12. financial results range

Observation. Among 75 independent financial_results events, the median session-close high-low range was 0.0319 (mean 0.0397, n=75), compared with 0.0148 for capital_structure.

Why it matters. Result days show a wider same-session price range than routine capital-structure notices.

Limitation. Range is window high / window low - 1. It is not a formal volatility model and does not isolate overnight gaps.

13. financial results session close

Observation. Among 75 independent financial_results events, the mean session-close return was -0.0050 and the median was -0.0063 (n=75). The 95% interval was [-0.0103, 0.0003].

Why it matters. The same-session financial-result return is the conditioning input for PEAD, so its sign split matters more than the pooled mean.

Limitation. The interval includes values near zero. Mean and median agree on sign but the effect is small and is not a trading rule.

14. price conditioned pead

Observation. This is a price-conditioned analysis based on observed post-announcement price reaction, not an earnings-surprise analysis. Positive session-close reaction events (n=26) had mean D+1 0.0139 and mean D+3 0.0381. Negative reaction events (n=49) had mean D+1 -0.0217 and mean D+3 -0.0167. The unconditioned financial-result D+1 mean was -0.0094 (n=75). Neutral initial-reaction events had 0 observations.

Why it matters. The assignment requires price-conditioned PEAD on independent financial-result events. The sample shows later-session means that keep the sign of the event-day reaction.

Limitation. Stock-wise sign cells are mostly n < 10 and are marked descriptive. No benchmark residual is removed. This pattern is descriptive, not causal, and is not an earnings-surprise result.

15. capital structure cluster concentration

Observation. capital_structure has 577 announcements but only 300 independent events (announcement-to-cluster ratio 1.92). The largest cluster size is 5.

Why it matters. Treating every trading-window or allotment notice as independent would overstate the sample for this subject.

Limitation. Cluster membership is rule-based on normalized subject text and a 48-hour gap. Independence is an implementation definition, not a statistical guarantee.

16. business updates session close

Observation. Among 688 independent business_updates events, the mean session-close return was -0.0018 and the median was -0.0013 (n=687). The 95% interval was [-0.0032, -0.0003].

Why it matters. This is the largest subject group. The same-session mean is close to zero, which is useful context for the more active financial-result windows.

Limitation. business_updates is a residual taxonomy bucket and mixes many notice types. A single mean should not be read as a common economic mechanism. Intraday 60-minute missingness reaches 0.102 in some subjects.

17. stock wise pead dispersion

Observation. Stock-wise financial-result PEAD means differ by symbol. HAL D+1 mean=0.0123 (n=13); HDFCBANK D+1 mean=-0.0063 (n=13); NYKAA D+1 mean=-0.0250 (n=13); RELIANCE D+1 mean=-0.0033 (n=12); RVNL D+1 mean=-0.0173 (n=24). Sign-split stock cells are mostly n < 10.

Why it matters. A pooled PEAD average can hide stock-level differences in a five-name sample.

Limitation. Per-stock financial-result samples are 12 to 24 independent events. Conditioned stock cells are descriptive_only and should not be ranked as strategies.

9. Limitations

- No earnings-surprise data and no external analyst estimates are available.
- Timestamps stay naive because DATA_FORMAT.md was not in the assessment pack.
- Returns are raw close-to-close values. There is no benchmark residual or beta adjustment.
- Volume baselines use at most 20 prior sessions and need at least 5 comparable windows.
- Market coverage ends in the supplied tape, so late events lose D+10 and D+20.
- Stock-subject and stock-condition PEAD cells are often small and descriptive_only.
- Confidence intervals are simple normal-approximation intervals and are not hypothesis tests.
- Clustering and taxonomy are deterministic rules. They can miss economic links that text rules do not see.
- Price-conditioned PEAD describes later returns after an observed price reaction. It is not causal evidence.
- Two announcements are excluded. Two invalid OHLC bars exist and are not repaired.

10. Reproducibility

Create a virtual environment, install requirements.txt, and pass the supplied assessment folder at run time. Do not hard-code a machine path. Commands: `python -m src.run_audit`, `python -m src.run_taxonomy`, `python -m src.run_event_impact`, `python -m src.run_pead`, `python -m src.run_subject_analysis`, `python -m src.run_final_evidence`, `python -m src.run_report`. Tests: `python -m pytest -q`. Raw announcement and market CSVs stay outside the public repository. Generated CSVs, figures, and this report are written under outputs/.

Conclusion

In this five-stock sample, financial-result events are the noisiest same-session windows on volume and range. Pooled same-session returns are small. When financial-result events are split by the observed session-close price reaction, later-session means keep that sign. The pattern is descriptive, price-conditioned, and limited by sample size, the absence of surprise data, and the lack of a benchmark adjustment. The analysis does not support a trading strategy claim.